

Problem Set 4

Seung Hyun Kim

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Problem 1

1.1 1-(a)

By a north star, the authors refer to a common goal that a field can collectively work toward achieving. They believe that having the north star of computer vision, or object recognition, has been tremendously useful for the development of machine learning as a way to focus the efforts of computer scientists into making enormous strides in one crucial area. Among other things, the existence of this particular north star has allowed: neural networks to be discovered as the workhorse of machine learning algorithms; and made possible revolutions in other related areas such as object detection.

1.2 1-(b)

A promising north star for macroeconomics seems to be a global solution for heterogeneous agent models with aggregate risk. So far, the literature has been devoted mostly to approximate first order solutions around the steady state, with rarely any attempts made toward searching for a global solution. The lack of a global solution has spillover effects that stall macroeconomic research further. For instance, since there is no clear solution to these models, we cannot even begin to think about estimating them (Mikkel's words...), and it is not clear how to deal with large shocks such as the COVID shock in the context of first order perturbation methods. This suggests that macro would be better off if we could first find a universally agreed upon means of finding global solutions to heterogeneous agent models, and then proceeding from there.

Of course, I'm implicitly assuming that models with a continuum of heterogeneous agents subject to uninsurable idiosyncratic shocks and aggregate shocks will continue to dominate macroeconomics. I think it's as likely that another, more tractable class of models will be developed that preserve the core ideas of these heterogeneous agent models.

Problem 2

2.1 2-(a)

Let a_t^i indicate household I 's wealth at time t . Let θ_t^i be the share of wealth invested in capital and $1 - \theta_t^i$ the share invested in bonds. The output $y_{t,k}^i$ produced by capital follows the SDE

$$dy_{t,k}^i = \bar{R}k_t^i dt + \sigma k_t^i dW_t^i,$$

while the output $y_{t,b}^i$ produced by bonds follows

$$dy_{t,b}^i = r_t b_t^i dt,$$

where r_t is the instantaneous return to holding a unit of the risk-free bond. It follows that wealth evolves according to the SDE

$$\begin{aligned} da_t^i &= dy_{t,k}^i + dy_{t,b}^i - c_t dt \\ &= (\bar{R}k_t^i + r_t b_t^i - c_t) dt + \sigma k_t^i dW_t^i \\ &= (\bar{R}\theta_t^i a_t^i + r_t(1 - \theta_t^i)a_t^i - c_t) dt + \sigma \theta_t^i a_t^i dW_t^i \\ &= \left[\left(r_t + (\bar{R} - r_t)\theta_t^i \right) a_t^i - c_t \right] dt + \sigma \theta_t^i a_t^i dW_t^i. \end{aligned}$$

Household i now solves the sequence problem

$$\begin{aligned} \max_{(c,\theta) \in \mathcal{C}(a_0)} \quad & \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} u(c_t^i) ds \right] \\ \text{subject to} \quad & da_t^i = \left[\left(r_t + (\bar{R} - r_t)\theta_t^i \right) a_t^i - c_t^i \right] dt + \sigma \theta_t^i a_t^i dW_t^i \\ & a_0^i = a_0, \end{aligned}$$

where $\mathcal{C}(a_0)$ collects the paths $\{c_t, \theta_t\}_{t \in \mathbb{N}}$ of consumption and capital shares such that the objective function converges.

The HJBE corresponding to this problem is given as

$$\rho V_t(a) = \max_{c,\theta} \left[u(c) + \partial_a V_t(a) \left((r_t + (\bar{R} - r_t)\theta)a - c \right) + \frac{1}{2} \sigma^2 \theta^2 a^2 \partial_{aa} V_t(a) \right] + \partial_t V_t(a).$$

Suppose that the solution to this problem is time-invariant, that is, $V_t(a) = V(a)$ and $r_t = r$ for any $t \in \mathbb{N}$. Then, we obtain the stationary HJBE

$$\rho V(a) = \max_{c, \theta} \left[u(c) + \partial_a V(a) \left((r + (\bar{R} - r)\theta)a - c \right) + \frac{1}{2} \sigma^2 \theta^2 a^2 \partial_{aa} V(a) \right].$$

The first order conditions for maximization are given as

$$u'(c^*(a)) = \partial_a V(a),$$

$$\partial_a V(a)(\bar{R} - r)a + \left(\sigma^2 a^2 \partial_{aa} V(a) \right) \theta^*(a) = 0,$$

so that

$$c^*(a) = (\partial_a V(a))^{-\frac{1}{\gamma}}$$

$$\theta^*(a) = -\frac{\bar{R} - r}{\sigma^2} \left(\frac{\partial_a V(a)}{a \partial_{aa} V(a)} \right)$$

in light of the CRRA form $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ for utility.

Here we make the guess that $V(a)$ is given in the same CRRA form as flow utility, that is,

$$V(a) = \psi \frac{a^{1-\gamma}}{1-\gamma}$$

for some constant ψ . It follows then that

$$-\frac{\partial_a V(a)}{a \partial_{aa} V(a)} = \frac{1}{\gamma},$$

so that the optimal share of capital is given as a simple function that is increasing in the risk premium $\bar{R} - r$ and decreasing in the volatility σ^2 of capital returns:

$$\theta^* = \frac{\bar{R} - r}{\gamma \sigma^2}.$$

Furthermore, the first order condition for consumption allows us to solve for optimal consumption as

$$c^*(a) = \psi^{-\frac{1}{\gamma}} a.$$

Finally, since the HJBE tells us that

$$\begin{aligned} \rho \psi \frac{a^{1-\gamma}}{1-\gamma} &= \rho V(a) = u(c^*(a)) + \partial_a V(a) \left[\left(r + (\bar{R} - r)\theta^* \right) a - c^*(a) \right] + \frac{1}{2} \sigma^2 (\theta^*)^2 a^2 \partial_{aa} V(a) \\ &= \frac{1}{1-\gamma} \psi^{-\frac{1-\gamma}{\gamma}} a^{1-\gamma} + \psi a^{1-\gamma} \left[\left(r + \frac{(\bar{R} - r)^2}{\sigma^2 \gamma} \right) - \psi^{-\frac{1}{\gamma}} \right] - \psi \frac{1}{2} \frac{(\bar{R} - r)^2}{\sigma^2 \gamma} a^{1-\gamma}, \end{aligned}$$

dividing both sides by $V(a)$ shows us that

$$\begin{aligned}\rho &= \psi^{-\frac{1}{\gamma}} + (1-\gamma) \left[\left(r + \frac{(\bar{R}-r)^2}{\sigma^2\gamma} \right) - \psi^{-\frac{1}{\gamma}} \right] - (1-\gamma) \frac{1}{2} \frac{(\bar{R}-r)^2}{\sigma^2\gamma} \\ &= \gamma \psi^{-\frac{1}{\gamma}} + (1-\gamma) \left(r + \frac{1}{2} \frac{(\bar{R}-r)^2}{\sigma^2\gamma} \right).\end{aligned}$$

Solving for ψ gives us

$$\psi^{-\frac{1}{\gamma}} = \frac{1}{\gamma} \left[\rho - (1-\gamma) \left(r + \frac{1}{2} \frac{(\bar{R}-r)^2}{\sigma^2\gamma} \right) \right].$$

It remains to solve for the interest rate r , which can be done by studying the bond market clearing condition

$$(1-\theta^*) \int_0^1 a_t^i di = \int_0^1 b_t^i di = 0.$$

This shows us that $\theta^* = 1$; otherwise, $\int_0^1 a_t^i di = 0$ and there is no aggregate capital in the economy, which in turn means that there is no output and no consumption, which violates the optimality condition of the household. In other words, market clearing requires

$$\frac{\bar{R}-r}{\gamma\sigma^2} = 1,$$

so that the equilibrium interest rate is

$$r = \bar{R} - \gamma\sigma^2.$$

In summary, the value function and interest rate at stationary recursive competitive equilibrium are given uniquely as

$$\begin{aligned}V(a) &= \left(\frac{1}{\gamma} \left[\rho - (1-\gamma) \left(\bar{R} - \gamma\sigma^2 + \frac{\gamma}{2\sigma^2} \right) \right] \right)^{-\gamma} \frac{a^{1-\gamma}}{1-\gamma} \\ r &= \bar{R} - \gamma\sigma^2.\end{aligned}$$

2.2 2-(b)

We are able to decouple the HJBE from the KFE because, in this setup, every household chooses to save the same share θ^* in capital. This means that this particular heterogeneous agent model has a market clearing condition that is exactly that of a representative agent model, which allows us to solve for the equilibrium interest rate even without knowing the distribution of wealth in the economy.

If there are borrowing constraints, we now need to know or at least simulate the wealth distribution to obtain the equilibrium interest rate. This is because households near the borrowing constraint will hold a different share of their wealth in capital compared to households with higher wealth, meaning $\theta^*(a)$ now depends on a .

2.3 2-(c)

Using the equilibrium interest rates and optimal consumption/portfolio shares derived earlier, at equilibrium household i 's wealth a_t^i evolves according to the diffusion

$$da_t^i = \left(\bar{R} - \psi^{-\frac{1}{\gamma}}\right) a_t^i dt + \sigma a_t^i dW_t^i$$

It follows that household i 's wealth follows a geometric Brownian motion, so that

$$d\log(a_t^i) = \left(\bar{R} - \psi^{-\frac{1}{\gamma}} - \frac{\sigma^2}{2}\right) dt + \sigma dW_t^i$$

and we have

$$a_t^i = a_0^i e^{\left(\bar{R} - \psi^{-\frac{1}{\gamma}} - \frac{\sigma^2}{2}\right)t + \sigma W_t^i}.$$

It follows that wealth follows a lognormal distribution with mean $\left(\bar{R} - \psi^{-\frac{1}{\gamma}} - \frac{\sigma^2}{2}\right)t$ and variance σ^2 . Unless $\bar{R} - \psi^{-\frac{1}{\gamma}} - \frac{\sigma^2}{2}$ is exactly equal to 0, the problem does not admit a stationary wealth distribution.

Problem 3

3.1 The Model: Firm Decisions and Market Clearing

In the KS model, there exists a continuum of households indexed by $i \in [0, 1]$ that faces idiosyncratic labor supply and aggregate productivity shocks. Households receive labor income and save in a short term bond with wage w_t and interest rate r_t . They also face a borrowing constraint, so that the asset state space is given as $\mathcal{A} = [\underline{a}, +\infty)$. Labor supply follows a Poisson jump process and takes values in $\mathcal{N} = \{n_1, n_2\}$ with intensity matrix

$$\Lambda = \begin{pmatrix} -\lambda(n_1) & \lambda(n_1) \\ \lambda(n_2) & -\lambda(n_2) \end{pmatrix}.$$

Log TFP follows the OU process

$$dZ_t = \underbrace{\eta(\bar{Z} - Z_t)}_{\mu^z(Z_t)} dt + \sigma dW_t^0.$$

The representative firm faces a Cobb-Douglas production function with capital and labor, and the depreciation rate is $\delta \in (0, 1)$, so that the firm's profit maximization problem at time t is

$$\max_{K_t, L_t} \left[e^{Z_t} K_t^\alpha L_t^{1-\alpha} - (r_t + \delta) K_t - w_t L_t \right].$$

The first order conditions for maximization are given as

$$r_t = \alpha e^{Z_t} K_t^{\alpha-1} L_t^{1-\alpha} - \delta,$$

$$w_t = (1 - \alpha) e^{Z_t} K_t^\alpha L_t^{-\alpha}.$$

Let $g_t(\cdot)$ be the density of wealth and labor supply at equilibrium. The market clearing conditions for the capital and labor markets are

$$L_t = \sum_{j=1}^2 n_j \left(\int_0^1 g_t(a, n_j) da \right)$$

$$K_t = \sum_{j=1}^2 \left(\int_0^1 a g_t(a, n_j) da \right).$$

where $g_t(\cdot)$ is the conditional distribution of (a_t^i, n_t^i) given the path of the aggregate shocks. The goods market clearing conditions are satisfied by Walras' law.

3.2 The Model: The Household HJBE

Combining the market clearing conditions with the firm's profit maximization problems, we can express the prices $q_t = (r_t, w_t)'$ solely as functions of the aggregate shock Z_t and the distribution g_t :

$$q_t = Q(Z_t, g_t) = \begin{pmatrix} r(Z_t, g_t) \\ w(Z_t, g_t) \end{pmatrix}.$$

Suppose household i believes that the distribution g_t evolves according to the KFE

$$dg_t(a, n) = \tilde{\mu}^g(a, n; Z_t, g_t)dt.$$

Household i then solves the maximization problem

$$\begin{aligned} & \max_{\{c_t^i\}_{t \in \mathbb{N}}} \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} u(c_t^i) dt \right] \\ \text{subject to } & d \begin{bmatrix} a_t^i \\ n_t^i \\ Z_t \end{bmatrix} = \begin{bmatrix} w(Z_t, g_t)n_t^i + r(Z_t, g_t)a_t^i - c_t^i \\ 0 \\ \mu^z(Z_t) \end{bmatrix} dt + \begin{bmatrix} 0 \\ \tilde{n}_t^i - n_t^i \\ 0 \end{bmatrix} dN_t^i + \begin{bmatrix} 0 \\ 0 \\ \sigma \end{bmatrix} dW_t^0 \\ & dg_t(a, n) = \tilde{\mu}^g(a, n; Z_t, g_t)dt \quad \forall (a, n) \in \mathcal{A} \times \mathcal{N} \\ & a_t \geq \underline{a}. \end{aligned}$$

This problem has four states: wealth a , labor supply n , aggregate shocks z and the wealth distribution g . We are interested in the HJBE that characterizes the stationary value function $V(a, n, z, g)$. Below we (roughly) derive it from first principles.

3.2.1 HJBE Derivation: Ito's Lemma for Drift Jump Diffusion Part

Suppose that the value function at time 0 is already equal to the steady state value function, and let $\mathcal{C}^*(a, n, z, g)$ be the optimal path of consumption given initial values a, n, z, g of the state

variables. For any small time increment $\Delta > 0$, let $\{c_t^i\}_{t \geq 0}$ be a stream of consumption satisfying

$$\{c_t^i\}_{t \geq \Delta} = C^*(a_\Delta, n_\Delta, Z_\Delta, g_\Delta).$$

Defining

$$\bar{V}_t(a, n, z, g) = e^{-\rho t} V(a, n, z, g),$$

we have

$$\begin{aligned} \bar{V}_0(a, n, z, g) &\geq \mathbb{E}_0 \left[\int_0^\Delta e^{-\rho t} u(c_t^i) dt \right] + \mathbb{E}_0 \left[\mathbb{E}_\Delta \left[\int_\Delta^\infty e^{-\rho t} u(c_t^i) dt \right] \right] \\ &\geq \mathbb{E}_0 \left[\int_0^\Delta e^{-\rho t} u(c_t^i) dt \right] + \mathbb{E}_0 \left[\bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g_\Delta) \right], \end{aligned}$$

so that

$$0 \geq \mathbb{E}_0 \left[\int_0^\Delta e^{-\rho t} u(c_t^i) dt \right] + \mathbb{E}_0 \left[\bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g_\Delta) - \bar{V}_0(a, n, z, g) \right].$$

We now apply Ito's lemma to the second term as follows. First, note that

$$\begin{aligned} &\frac{\bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g_\Delta) - \bar{V}_0(a, n, z, g)}{\Delta} \\ &= \frac{\bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g) - \bar{V}_0(a, n, z, g)}{\Delta} + \frac{\bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g_\Delta) - \bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g)}{\Delta}. \end{aligned}$$

Applying Ito's lemma to the mapping $(t, a, n, z) \mapsto \bar{V}_t(a, n, z, g)$ reveals that the first term can be written as

$$\begin{aligned} &\frac{\bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g) - \bar{V}_0(a, n, z, g)}{\Delta} \\ &= \frac{1}{\Delta} \int_0^\Delta \left(\partial_t \bar{V}_t(a_t, n_t, z_t, g) + \partial_a \bar{V}_t(a_t, n_t, z_t, g) [w(z_t, g)n_t + r(z_t, g)a_t - c_t^i] \right. \\ &\quad \left. + \partial_z \bar{V}_t(a_t, n_t, z_t, g) \mu_z(z_t) + \frac{1}{2} \partial_{zz} \bar{V}_t(a_t, n_t, z_t, g) \sigma^2 \right) dt \\ &\quad + \sigma \frac{1}{\Delta} \int_0^\Delta \partial_z \bar{V}_t(a_t, n_t, z_t, g) dW_t^0 + \frac{1}{\Delta} \int_0^\Delta \left(\bar{V}_t(a_t, \check{n}_t, z_t, g) - \bar{V}_t(a_t, n_t, z_t, g) \right) dN_t^i, \end{aligned}$$

and taking time 0 expectations on both sides yields

$$\begin{aligned} &\frac{1}{\Delta} \mathbb{E}_0 \left[\bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g) - \bar{V}_0(a, n, z, g) \right] \\ &= \frac{1}{\Delta} \int_0^\Delta \mathbb{E}_0 \left[\partial_t \bar{V}_t(a_t, n_t, z_t, g) + \partial_a \bar{V}_t(a_t, n_t, z_t, g) [w(z_t, g)n_t + r(z_t, g)a_t - c_t^i] \right] \end{aligned}$$

$$\begin{aligned}
& + \partial_z \bar{V}_t(a_t, n_t, z_t, g) \mu_z(z_t) + \frac{1}{2} \partial_{zz} \bar{V}_t(a_t, n_t, z_t, g) \sigma^2 \Big] dt \\
& + \int_0^\Delta \mathbb{E}_0 \left[\lambda(n_t) \left(\bar{V}_t(a_t, \check{n}_t, z_t, g) - \bar{V}_t(a_t, n_t, z_t, g) \right) \right] dt.
\end{aligned}$$

Finally, taking $\Delta \rightarrow 0$ yields, by the FTC,

$$\begin{aligned}
& \lim_{\Delta \rightarrow 0} \frac{\mathbb{E}_0 \left[\bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g) - \bar{V}_0(a, n, z, g) \right]}{\Delta} \\
& = -\rho V(a, n, z, g) + \partial_a V(a, n, z, g) \left[w(z, g)n + r(z, g)a - c_0^i \right] \\
& \quad + \partial_z V(a, n, z, g) \mu_z(z) + \frac{1}{2} \sigma^2 \partial_{zz} V(a, n, z, g) \\
& \quad + \lambda(n) \left(V(a, \check{n}, z, g) - V(a, n, z, g) \right).
\end{aligned}$$

3.2.2 HJBE Derivation: Infinite-Dimensional Part

Moving onto the second term, let $\mathbf{m}$ be the product of the Lebesgue measure and the counting measure on $\mathcal{A} = [\underline{a}, +\infty)$ and $\mathcal{N} = \{n_1, n_2\}$. Then, $\mathbf{m}$ is a σ -finite measure on $\mathcal{A} \times \mathcal{N}$, and we let $L^2(\mathbf{m})$ denote the set of all functions on $\mathcal{A} \times \mathcal{N}$ that are square integrable with respect to $\mathbf{m}$. From the usual results on L^2 spaces we know that $\mathbf{X} = L^2(\mathbf{m})$ is a Hilbert space with inner product

$$\langle f, g \rangle_{\mathbf{X}} = \sum_{j=1}^2 \int_{\underline{a}}^{\infty} f(a, n_j) g(a, n_j) da$$

for any $f, g \in \mathbf{X}$.

We assume that $\{V(a, n, z, \cdot)\}_{(a, n, z)}$ is a collection of functions from $L^2(\mathbf{m})$ into $\mathbb{R}$ that are Frechet-differentiable, that is, we assume that, for any values (a, n, z) of the states excluding the wealth distribution and any $f \in L^2(\mathbf{m})$, there exists a continuous linear operator $DV(a, n, z, f) : L^2(\mathbf{m}) \rightarrow \mathbb{R}$ such that

$$\lim_{\|h\|_{\mathbf{X}} \rightarrow 0} \frac{|V(a, n, z, f+h) - V(a, n, z, f) - [DV(a, n, z, f)](h)|}{\|h\|_{\mathbf{X}}} = 0.$$

Since $L^2(\mathbf{m})$ is a Hilbert space over the real field and $D_g V(a, n, z, g)$ is a continuous linear operator on that Hilbert space, the Riesz representation theorem tells us that there must exist some function $\frac{\partial V(a, n, z, f(\cdot))}{\partial f} \in L^2(\mathbf{m})$ such that

$$[D_g V(a, n, z, g)](h) = \left\langle \frac{\partial V(a, n, z, f(\cdot))}{\partial f}, h \right\rangle_{\mathbf{X}} = \sum_{j=1}^2 \int_{\underline{a}}^{\infty} \frac{\partial V(a, n, z, f(a', n_j))}{\partial f} h(a', n_j) da'$$

for any $h \in L^2(\mathbf{m})$. To deal with the stochasticity of the paths $\{a_t, n_t, z_t, g_t\}_{t \leq \Delta}$, we make the following assumptions:

- **Mean Square Frechet Differentiability**

For any sequence of random functions $\{h_\Delta\}$ taking values in $L^2(\mathbf{m})$ such that

$$\|h_\Delta\|_{\mathbf{X}} \xrightarrow{L^2} 0,$$

we have

$$\lim_{\Delta \rightarrow 0} \mathbb{E}_0 \left[\frac{|V(a_\Delta, n_\Delta, z_\Delta, g + h_\Delta) - V(a_\Delta, n_\Delta, z_\Delta, g) - [DV(a_\Delta, n_\Delta, z_\Delta, g)](h_\Delta)|^2}{\|h_\Delta\|_{\mathbf{X}}^2} \right] = 0.$$

- **Mean Square Continuity of Value Function**

As $\Delta \rightarrow 0$,

$$\left\| \frac{\partial V(a_\Delta, n_\Delta, z_\Delta, g(\cdot))}{\partial f} - \frac{\partial V(a, n, z, g(\cdot))}{\partial f} \right\|_{\mathbf{X}} \xrightarrow{L^2} 0.$$

- **FTC for Distribution Drift**

$\tilde{\mu}^g(\cdot; z, g)$ is square integrable with respect to $\mathbf{m}$ for any value of (z, g) and

$$\lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \int_0^\Delta \mathbb{E}_0 \left[\int_{\mathbf{X}} \tilde{\mu}^g(\cdot; z_t, g_t)^2 d\mathbf{m} \right] dt = \int_{\mathbf{X}} \tilde{\mu}^g(\cdot; z, g)^2 d\mathbf{m} < +\infty$$

$$\lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \int_0^\Delta \mathbb{E}_0 \left[\int_{\mathcal{A} \times \mathcal{N}} \frac{\partial V(a, n, z, g(\cdot))}{\partial f} \tilde{\mu}^g(\cdot; z_t, g_t) d\mathbf{m} \right] dt = \int_{\mathcal{A} \times \mathcal{N}} \frac{\partial V(a, n, z, g(\cdot))}{\partial f} \tilde{\mu}^g(\cdot; z, g) d\mathbf{m}.$$

Returning to the problem at hand, since

$$g_\Delta(a', n') - g(a', n') = \int_0^\Delta \tilde{\mu}^g(a', n'; z_t, g_t) dt$$

for any $(a', n') \in \mathcal{A} \times \mathcal{N}$, defining the function $h \in L^2(\mathbf{m})$ as

$$h_\Delta = \int_0^\Delta \tilde{\mu}^g(\cdot; z_t, g_t) dt$$

shows us that

$$\begin{aligned} \frac{\bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g_\Delta) - \bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g)}{\Delta} &= \frac{1}{\Delta} e^{-\rho\Delta} [V(a_\Delta, n_\Delta, Z_\Delta, g + h_\Delta) - V(a_\Delta, n_\Delta, Z_\Delta, g)] \\ &= \frac{1}{\Delta} e^{-\rho\Delta} [DV(a_\Delta, n_\Delta, z_\Delta, g)](h_\Delta) + R(\Delta), \end{aligned}$$

where

$$R(\Delta) = e^{-\rho\Delta} \frac{V(a_\Delta, n_\Delta, Z_\Delta, g + h_\Delta) - V(a_\Delta, n_\Delta, Z_\Delta, g) - [DV(a_\Delta, n_\Delta, z_\Delta, g)](h_\Delta)}{\Delta}.$$

Because

$$\begin{aligned} \|h_\Delta\|_{\mathbf{X}}^2 &= \int_{\mathbf{X}} \left| \int_0^\Delta \tilde{\mu}^g(\cdot; z_t, g_t) dt \right|^2 d\mathbf{m} \\ &\leq \Delta \int_{\mathbf{X}} \int_0^\Delta \tilde{\mu}^g(\cdot; z_t, g_t)^2 dt d\mathbf{m} && \text{(Hölder's Inequality)} \\ &= \Delta \int_0^\Delta \left(\int_{\mathbf{X}} \tilde{\mu}^g(\cdot; z_t, g_t)^2 d\mathbf{m} \right) dt, && \text{(Fubini's Theorem)} \end{aligned}$$

it follows that

$$\mathbb{E}_0 \left(\frac{\|h_\Delta\|_{\mathbf{X}}}{\Delta} \right)^2 \leq \frac{1}{\Delta} \int_0^\Delta \mathbb{E}_0 \left[\int_{\mathbf{X}} \tilde{\mu}^g(\cdot; z_t, g_t)^2 d\mathbf{m} \right] dt.$$

This result combined with Hölder's inequality shows us that

$$\begin{aligned} \mathbb{E}_0 [R(\Delta)] &\leq \left(\mathbb{E}_0 \left[\frac{|V(a_\Delta, n_\Delta, z_\Delta, g + h_\Delta) - V(a_\Delta, n_\Delta, z_\Delta, g) - [DV(a_\Delta, n_\Delta, z_\Delta, g)](h_\Delta)|^2}{\|h_\Delta\|_{\mathbf{X}}^2} \right] \right)^{\frac{1}{2}} \\ &\quad \times \left(\frac{1}{\Delta} \int_0^\Delta \mathbb{E}_0 \left[\int_{\mathbf{X}} \tilde{\mu}^g(\cdot; z_t, g_t)^2 d\mathbf{m} \right] dt \right)^{\frac{1}{2}} e^{-\rho\Delta}. \end{aligned}$$

By mean square Frechet differentiability the first term goes to 0 while the other terms are $O(1)$ as $\Delta \rightarrow 0$. As such,

$$\lim_{\Delta \rightarrow 0} \mathbb{E}_0 [R(\Delta)] = 0.$$

Meanwhile,

$$\begin{aligned} \frac{1}{\Delta} e^{-\rho\Delta} [DV(a_\Delta, n_\Delta, z_\Delta, g)](h_\Delta) &= \frac{1}{\Delta} e^{-\rho\Delta} \sum_{j=1}^2 \int_{\underline{a}}^\infty \frac{\partial V(a_\Delta, n_\Delta, z_\Delta, g(a', n_j))}{\partial f} \left(\int_0^\Delta \tilde{\mu}^g(a', n_j; z_t, g_t) dt \right) da' \\ &= e^{-\rho\Delta} \frac{1}{\Delta} \int_0^\Delta \sum_{j=1}^2 \left(\int_{\underline{a}}^\infty \frac{\partial V(a_\Delta, n_\Delta, z_\Delta, g(a', n_j))}{\partial f} \tilde{\mu}^g(a', n_j; z_t, g_t) da' \right) dt. \end{aligned}$$

This term can further be decomposed as

$$\underbrace{\frac{1}{\Delta} \int_0^\Delta \sum_{j=1}^2 \left(\int_{\underline{a}}^\infty \frac{\partial V(a, n, z, g(a', n_j))}{\partial f} \tilde{\mu}^g(a', n_j; z_t, g_t) da' \right) dt}_I$$

$$+ \underbrace{\frac{1}{\Delta} \int_0^\Delta \sum_{j=1}^2 \left(\int_a^\infty \left[\frac{\partial V(a, n, z, g(a', n_j))}{\partial f} - \frac{\partial V(a_\Delta, n_\Delta, z_\Delta, g(a', n_j))}{\partial f} \right] \tilde{\mu}^g(a', n_j; z_t, g_t) da' \right) dt}_{II},$$

where the second term is bounded above by

$$\frac{1}{\Delta} \left\| \frac{\partial V(a, n, z, g(\cdot))}{\partial f} - \frac{\partial V(a_\Delta, n_\Delta, z_\Delta, g(\cdot))}{\partial f} \right\|_{\mathbf{X}} \left[\int_0^\Delta \left(\int_{\mathcal{A} \times \mathcal{N}} \tilde{\mu}^g(\cdot; z_t, g_t)^2 d\mathbf{m} \right) dt \right]^{\frac{1}{2}}$$

by Hölder's inequality. Another application of Hölder's inequality shows us that this term is bounded above itself by

$$\frac{1}{\sqrt{\Delta}} \left\| \frac{\partial V(a, n, z, g(\cdot))}{\partial f} - \frac{\partial V(a_\Delta, n_\Delta, z_\Delta, g(\cdot))}{\partial f} \right\|_{\mathbf{X}} \left[\int_0^\Delta \left(\int_{\mathcal{A} \times \mathcal{N}} \tilde{\mu}^g(\cdot; z_t, g_t)^2 d\mathbf{m} \right) dt \right]^{\frac{1}{2}}.$$

Finally, taking expectations and using Hölder's inequality one last time shows us that

$$\begin{aligned} \mathbb{E}_0 |II| &\leq \left(\mathbb{E}_0 \left\| \frac{\partial V(a, n, z, g(\cdot))}{\partial f} - \frac{\partial V(a_\Delta, n_\Delta, z_\Delta, g(\cdot))}{\partial f} \right\|_{\mathbf{X}}^2 \right)^{\frac{1}{2}} \\ &\quad \times \left(\frac{1}{\Delta} \int_0^\Delta \mathbb{E}_0 \left[\int_{\mathcal{A} \times \mathcal{N}} \tilde{\mu}^g(\cdot; z_t, g_t)^2 d\mathbf{m} \right] dt \right)^{\frac{1}{2}}. \end{aligned}$$

The second term is $O(1)$ and the first term goes to 0 as $\Delta \rightarrow 0$ by assumption, so

$$\lim_{\Delta \rightarrow 0} \mathbb{E}_0 |II| = 0.$$

Finally,

$$\begin{aligned} \mathbb{E}_0 [I] &= \frac{1}{\Delta} \int_0^\Delta \mathbb{E}_0 \left[\int_{\mathcal{A} \times \mathcal{N}} \frac{\partial V(a, n, z, g(\cdot))}{\partial f} \tilde{\mu}^g(\cdot; z_t, g_t) d\mathbf{m} \right] dt \\ &\rightarrow \int_{\mathcal{A} \times \mathcal{N}} \frac{\partial V(a, n, z, g(\cdot))}{\partial f} \tilde{\mu}^g(\cdot; z, g) d\mathbf{m} = \sum_{j=1}^2 \int_a^\infty \frac{\partial V(a, n, z, g(a', n_j))}{\partial f} \tilde{\mu}^g(a', n_j; z, g) da' \end{aligned}$$

by the FTC assumption. Putting all this together shows us that

$$\lim_{\Delta \rightarrow 0} \frac{\mathbb{E}_0 [\bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g_\Delta) - \bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g)]}{\Delta} = \sum_{j=1}^2 \int_a^\infty \frac{\partial V(a, n, z, g(a', n_j))}{\partial f} \tilde{\mu}^g(a', n_j; z, g) da',$$

and therefore

$$\begin{aligned} &\frac{1}{\Delta} \mathbb{E}_0 [\bar{V}_\Delta(a_\Delta, n_\Delta, Z_\Delta, g_\Delta) - \bar{V}_0(a, n, z, g)] \\ &\rightarrow -\rho V(a, n, z, g) + \partial_a V(a, n, z, g) [w(z, g)n + r(z, g)a - c_0^i] \\ &\quad + \partial_z V(a, n, z, g) \mu_z(z) + \frac{1}{2} \sigma^2 \partial_{zz} V(a, n, z, g) \end{aligned}$$

$$+ \lambda(n) \left(V(a, \check{n}, z, g) - V(a, n, z, g) \right) \\ + \sum_{j=1}^2 \int_{\underline{a}}^{\infty} \frac{\partial V(a, n, z, g(a', n_j))}{\partial f} \tilde{\mu}^g(a', n_j; z, g) da'$$

as $\Delta \rightarrow 0$.

3.2.3 HJBE Derivation: The Final Step

Meanwhile,

$$\lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \mathbb{E}_0 \left[\int_0^{\Delta} e^{-\rho t} u(c_t^i) dt \right] = u(c_0^i)$$

by the FTC. Since

$$0 \geq \frac{1}{\Delta} \mathbb{E}_0 \left[\int_0^{\Delta} e^{-\rho t} u(c_t^i) dt \right] + \frac{1}{\Delta} \mathbb{E}_0 \left[\bar{V}_{\Delta}(a_{\Delta}, n_{\Delta}, Z_{\Delta}, g_{\Delta}) - \bar{V}_0(a, n, z, g) \right]$$

for any $\Delta > 0$, it follows that

$$0 \geq u(c_0^i) - \rho V(a, n, z, g) + \partial_a V(a, n, z, g) \left[w(z, g)n + r(z, g)a - c_0^i \right] \\ + \partial_z V(a, n, z, g) \mu_z(z) + \frac{1}{2} \sigma^2 \partial_{zz} V(a, n, z, g) \\ + \lambda(n) \left(V(a, \check{n}, z, g) - V(a, n, z, g) \right) \\ + \sum_{j=1}^2 \int_{\underline{a}}^{\infty} \frac{\partial V(a, n, z, g(a', n_j))}{\partial f} \tilde{\mu}^g(a', n_j; z, g) da',$$

rearranging which we have

$$\rho V(a, n, z, g) \geq u(c_0^i) + \partial_a V(a, n, z, g) \left[w(z, g)n + r(z, g)a - c_0^i \right] \\ + \partial_z V(a, n, z, g) \mu_z(z) + \frac{1}{2} \sigma^2 \partial_{zz} V(a, n, z, g) \\ + \lambda(n) \left(V(a, \check{n}, z, g) - V(a, n, z, g) \right) \\ + \sum_{j=1}^2 \int_{\underline{a}}^{\infty} \frac{\partial V(a, n, z, g(a', n_j))}{\partial f} \tilde{\mu}^g(a', n_j; z, g) da'.$$

Repeating the same procedure for the optimal path $\{c_t^i\}_{t \geq 0} = \mathcal{C}^*(a, n, z, g)$ now shows us that the above holds as an equality with the optimal time 0 consumption, which gives us precisely the HJBE we want:

$$\rho V(a, n, z, g) = \max_{c \geq 0} \left\{ u(c) + \partial_a V(a, n, z, g) \left[w(z, g)n + r(z, g)a - c \right] \right.$$

$$\begin{aligned}
& + \partial_z V(a, n, z, g) \mu_z(z) + \frac{1}{2} \sigma^2 \partial_{zz} V(a, n, z, g) \\
& + \lambda(n) \left(V(a, \check{n}, z, g) - V(a, n, z, g) \right) \\
& + \sum_{j=1}^2 \int_{\underline{a}}^{\infty} \frac{\partial V(a, n, z, g(a', n_j))}{\partial f} \tilde{\mu}^g(a', n_j; z, g) da' \Big\}
\end{aligned}$$

together with the boundary constraint

$$\partial_a V(\underline{a}, n, z, g) \geq u'(w(z, g)n + r(z, g)\underline{a}).$$

3.3 The Model: KFE

Now that we have our HJBE, we can see that optimal consumption is characterized by the FOC

$$u'[c^*(a, n, z, g)] = V_a(a, n, z, g).$$

This allows us to obtain the KFE, which we again derive from first principles. Let $\mathcal{F}_\infty$ be the σ -algebra representing the information in the path $\{Z_t\}_{t \in \mathbb{N}}$ of aggregate shocks. For any $t \geq 0$, $g_t(\cdot)$ is the density of X_t^i conditional on $\mathcal{F}_\infty$.

Note that the individual's state variable $X_t^i = (a_t^i, n_t^i)$ evolves according to the following jump diffusion:

$$dX_t^i = \begin{bmatrix} \mathbf{s}(a_t^i, n_t^i, Z_t, g_t; \tilde{\mu}^g) \\ 0 \end{bmatrix} dt + \begin{pmatrix} 0 \\ \check{n}_t^i - n_t^i \end{pmatrix} dN_t^i,$$

where

$$\mathbf{s}(a_t^i, n_t^i, Z_t; \tilde{\mu}^g) = w(Z_t, g_t)n + r(Z_t, g_t)a - c^*(a_t^i, n_t^i, Z_t, g_t; \tilde{\mu}^g)$$

and we make the dependence on the belief $dg_t = \tilde{\mu}^g(\cdot; Z_t, g_t)dt$ explicit.

3.3.1 KFE Derivation

As in the proof sketch of the KFE for a diffusion process, choose any smooth function $\phi(\cdot)$ on $\mathbb{R}_+ \times \mathbb{R}^2$. Suppose we have a sample $\{X_t^i\}_{i \in N_+}$ that are conditional independent given $\mathcal{F}_\infty$ and have conditional density $g_t(\cdot)$ given $\mathcal{F}_\infty$. For any $N \in N_+$, define

$$\bar{\phi}_{Nt} = \frac{1}{N} \sum_{i=1}^N \phi(t, X_t^i).$$

From Ito's lemma applied to jump diffusion processes we can see that

$$d\phi(t, X_t^i) = \left[\partial_t \phi(t, a_t^i, n_t^i) + \partial_a \phi(t, a_t^i, n_t^i) \mathbf{s}(a_t^i, n_t^i, Z_t, g_t) \right] dt + \left(\phi(t, a_t^i, \check{n}_t^i) - \phi(t, a_t^i, n_t^i) \right) dN_t^i,$$

so that

$$\begin{aligned} \bar{\phi}_{Nt} - \bar{\phi}_{N0} &= \frac{1}{N} \sum_{i=1}^N \left[\phi(t, X_t^i) - \phi(0, X_0^i) \right] \\ &= \frac{1}{N} \sum_{i=1}^N \int_0^t \left[\partial_s \phi(s, a_s^i, n_s^i) + \partial_a \phi(s, a_s^i, n_s^i) \mathbf{s}(a_s^i, n_s^i, Z_s, g_s) \right] ds \\ &\quad + \frac{1}{N} \sum_{i=1}^N \int_0^t \left(\phi(s, a_s^i, \check{n}_s^i) - \phi(s, a_s^i, n_s^i) \right) dN_s^i. \end{aligned}$$

By the WLLN, taking $N \rightarrow \infty$ yields

$$\bar{\phi}_{Nt} - \bar{\phi}_{N0} \xrightarrow{P} \mathbb{E} \left[\phi(t, X_t^i) - \phi(0, X_0^i) \mid \mathcal{F}_\infty \right] = \sum_{j=1}^2 \int_{\underline{a}}^\infty (\phi(t, a, n_j) - \phi(0, a, n_j)) g_t(a, n_j) da,$$

where the convergence is with respect to the underlying (unconditional) probability measure $\mathbb{P}$. Similarly,

$$\begin{aligned} \bar{\phi}_{Nt} - \bar{\phi}_{N0} &\xrightarrow{P} \mathbb{E} \left[\int_0^t \left(\partial_s \phi(s, a_s^i, n_s^i) + \partial_a \phi(s, a_s^i, n_s^i) \mathbf{s}(a_s^i, n_s^i, Z_s, g_s; \tilde{\mu}^g) \right) ds \mid \mathcal{F}_\infty \right] \\ &\quad + \mathbb{E} \left[\int_0^t \left(\phi(s, a_s^i, \check{n}_s^i) - \phi(s, a_s^i, n_s^i) \right) dN_s^i \mid \mathcal{F}_\infty \right] \\ &= \int_0^t \left[\sum_{j=1}^2 \int_{\underline{a}}^\infty \left(\partial_s \phi(s, a, n_j) + \partial_a \phi(s, a, n_j) \mathbf{s}(a, n_j, Z_s, g_s; \tilde{\mu}^g) \right) g_s(a, n_j) da \right] ds \\ &\quad + \int_0^t \left[\sum_{j=1}^2 \lambda(n_j) \int_{\underline{a}}^\infty (\phi(s, a, \check{n}_j) - \phi(s, a, n_j)) g_s(a, n_j) da \right] ds. \end{aligned}$$

It follows from the almost sure uniqueness of probability limits that

$$\begin{aligned} \sum_{j=1}^2 \int_{\underline{a}}^\infty (\phi(t, a, n_j) - \phi(0, a, n_j)) g_t(a, n_j) da \\ &= \int_0^t \left[\sum_{j=1}^2 \int_{\underline{a}}^\infty \partial_s \phi(s, a, n_j) g_s(a, n_j) da \right] ds \\ &\quad + \int_0^t \left[\sum_{j=1}^2 \int_{\underline{a}}^\infty \partial_a \phi(s, a, n_j) \mathbf{s}(a, n_j, Z_s, g_s; \tilde{\mu}^g) g_s(a, n_j) da \right] ds \\ &\quad + \int_0^t \left[\sum_{j=1}^2 \lambda(n_j) \int_{\underline{a}}^\infty (\phi(s, a, \check{n}_j) - \phi(s, a, n_j)) g_s(a, n_j) da \right] ds \end{aligned}$$

almost surely.

We study each term in turn. For the first term, using integration by parts with respect to time yields

$$\begin{aligned}
\int_0^t \left[\sum_{j=1}^2 \int_{\underline{a}}^{\infty} \partial_s \phi(s, a, n_j) g_s(a, n_j) da \right] ds &= \int_0^t \int_{\mathcal{A} \times \mathcal{N}} \partial_s \phi(s, x) g_s(x) d\mathbf{m}(x) ds \\
&= \int_{\mathcal{A} \times \mathcal{N}} \int_0^t \partial_s \phi(s, x) g_s(x) ds d\mathbf{m}(x) \\
&= \int_{\mathcal{A} \times \mathcal{N}} \left([\phi(s, x) g_s(x)]_0^t - \int_0^t \phi(s, x) \partial_s g_s(x) ds \right) d\mathbf{m}(x) \\
&= \sum_{j=1}^2 \int_{\underline{a}}^{\infty} (\phi(t, a, n_j) g_t(a, n_j) - \phi(0, a, n_j) g_0(a, n_j)) da \\
&\quad - \int_0^t \left[\sum_{j=1}^2 \int_{\underline{a}}^{\infty} \phi(s, a, n_j) \partial_s g_s(a, n_j) da \right] ds.
\end{aligned}$$

Meanwhile, using integration by parts with respect to wealth for the second term yields

$$\begin{aligned}
&\int_0^t \left[\sum_{j=1}^2 \int_{\underline{a}}^{\infty} \partial_a \phi(s, a, n_j) \mathbf{s}(a, n_j, Z_s, g_s; \tilde{\mu}^g) g_s(a, n_j) da \right] ds \\
&= \sum_{j=1}^2 \int_0^t \left([\phi(s, a, n_j) \mathbf{s}(a, n_j, Z_s, g_s; \tilde{\mu}^g) g_s(a, n_j)]_{\underline{a}}^{\infty} - \int_{\underline{a}}^{\infty} \phi(s, a, n_j) \partial_a [\mathbf{s}(a, n_j, Z_s, g_s; \tilde{\mu}^g) g_s(a, n_j)] da \right) ds \\
&= - \int_0^t \left(\sum_{j=1}^2 \int_{\underline{a}}^{\infty} \phi(s, a, n_j) \partial_a [\mathbf{s}(a, n_j, Z_s, g_s; \tilde{\mu}^g) g_s(a, n_j)] da \right) ds
\end{aligned}$$

since, for any $s \geq 0$, $\mathbf{s}(a, n_j, Z_s, g_s; \tilde{\mu}^g) = 0$ and $g_s(a, n_j) \rightarrow 0$ as $a \rightarrow \infty$ by Lebesgue integrability.

Finally, we can see that

$$\begin{aligned}
&\int_0^t \left[\sum_{j=1}^2 \lambda(n_j) \int_{\underline{a}}^{\infty} (\phi(s, a, \check{n}_j) - \phi(s, a, n_j)) g_s(a, n_j) da \right] ds \\
&= \int_0^t \left[\sum_{j=1}^2 \lambda(n_j) \int_0^t \int_{\underline{a}}^{\infty} \phi(s, a, \check{n}_j) g_s(a, n_j) da \right] ds - \int_0^t \left[\sum_{j=1}^2 \lambda(n_j) \int_0^t \int_{\underline{a}}^{\infty} \phi(s, a, n_j) g_s(a, n_j) da \right] ds \\
&= \int_0^t \left[\sum_{j=1}^2 \lambda(\check{n}_j) \int_0^t \int_{\underline{a}}^{\infty} \phi(s, a, n_j) g_s(a, \check{n}_j) da \right] ds - \int_0^t \left[\sum_{j=1}^2 \lambda(n_j) \int_0^t \int_{\underline{a}}^{\infty} \phi(s, a, n_j) g_s(a, n_j) da \right] ds \\
&= \int_0^t \left[\sum_{j=1}^2 \int_{\underline{a}}^{\infty} \phi(s, a, n_j) (\lambda(\check{n}_j) g_s(a, \check{n}_j) - \lambda(n_j) g_s(a, n_j)) da \right] ds.
\end{aligned}$$

Putting all these together shows us that

$$\begin{aligned}
& \sum_{j=1}^2 \int_{\underline{a}}^{\infty} (\phi(t, a, n_j) - \phi(0, a, n_j)) g_t(a, n_j) da \\
&= \sum_{j=1}^2 \int_{\underline{a}}^{\infty} (\phi(t, a, n_j) g_t(a, n_j) - \phi(0, a, n_j) g_0(a, n_j)) da \\
&\quad - \int_0^t \left[\sum_{j=1}^2 \int_{\underline{a}}^{\infty} \phi(s, a, n_j) \partial_s g_s(a, n_j) da \right] ds \\
&\quad - \int_0^t \left(\sum_{j=1}^2 \int_{\underline{a}}^{\infty} \phi(s, a, n_j) \partial_a \left[\mathbf{s}(a, n_j, Z_s, g_s; \tilde{\mu}^g) g_s(a, n_j) \right] \right) da \\
&\quad + \int_0^t \left[\sum_{j=1}^2 \int_{\underline{a}}^{\infty} \phi(s, a, n_j) (\lambda(\check{n}_j) g_s(a, \check{n}_j) - \lambda(n_j) g_s(a, n_j)) da \right] ds
\end{aligned}$$

almost surely, and rearranging terms reveals

$$\begin{aligned}
0 &= \int_0^t \sum_{j=1}^2 \int_{\underline{a}}^{\infty} \phi(s, a, n_j) \left(-\partial_s g_s(a, n_j) - \partial_a \left[\mathbf{s}(a, n_j, Z_s, g_s; \tilde{\mu}^g) g_s(a, n_j) \right] \right. \\
&\quad \left. + \lambda(\check{n}_j) g_s(a, \check{n}_j) - \lambda(n_j) g_s(a, n_j) \right) da ds
\end{aligned}$$

almost surely.

Given $t > 0$ and any point $(a_0, n_0) \in \mathcal{A} \times \mathcal{N}$, it is well-established that there exists a sequence of compactly supported smooth functions $\{\phi_k\}_{k \in N_+}$ on $\mathbb{R}_+ \times \mathbb{R}^2$ that converges to the point mass $I_{(t, a_0, n_0)}$ with respect to the Lebesgue L^2 -norm as $k \rightarrow \infty$. We established above that

$$\begin{aligned}
0 &= \int_0^t \sum_{j=1}^2 \int_{\underline{a}}^{\infty} \phi_k(s, a, n_j) \left(-\partial_s g_s(a, n_j) - \partial_a \left[\mathbf{s}(a, n_j, Z_s, g_s; \tilde{\mu}^g) g_s(a, n_j) \right] \right. \\
&\quad \left. + \lambda(\check{n}_j) g_s(a, \check{n}_j) - \lambda(n_j) g_s(a, n_j) \right) da ds
\end{aligned}$$

almost surely for each $k \in N_+$; by the countability of the index N_+ , this is equivalent to saying that the above equality holds under any $k \in N_+$ with probability 1. Taking $k \rightarrow \infty$ on both sides implies, by the dominated convergence theorem, that

$$\partial_t g_t(a_0, n_0) = -\partial_a \left[\mathbf{s}(a_0, n_0, Z_t, g_t; \tilde{\mu}^g) g_t(a_0, n_0) \right] + \lambda(\check{n}_0) g_t(a_0, \check{n}_0) - \lambda(n_0) g_t(a_0, n_0),$$

which is just the usual KFE.

3.4 The Model: Equilibrium Concept

Now we can characterize equilibrium in the economy via a single master equation. We define a fully recursive competitive equilibrium as the collection of:

- A value function $V(a, n, z, g)$,
- A belief $\tilde{\mu}^g$ about the drift of the wealth distribution,
- A consumption rule $c^*(a, n, z, g; \tilde{\mu}^g)$ and the corresponding saving rule $s(a, n, z, g; \tilde{\mu}^g)$, and
- Pricing functions $w(z, g)$ and $r(z, g)$

such that:

i) Household Optimality

The value function satisfies the HJBE

$$\begin{aligned} \rho V(a, n, z, g; \tilde{\mu}^g) = \max_{c \geq 0} & \left\{ u(c) + \partial_a V(a, n, z, g; \tilde{\mu}^g) [w(z, g)n + r(z, g)a - c] \right. \\ & + \partial_z V(a, n, z, g; \tilde{\mu}^g) \mu_z(z) + \frac{1}{2} \sigma^2 \partial_{zz} V(a, n, z, g; \tilde{\mu}^g) \\ & + \lambda(n) \left(V(a, \tilde{n}, z, g; \tilde{\mu}^g) - V(a, n, z, g; \tilde{\mu}^g) \right) \\ & \left. + \sum_{j=1}^2 \int_{\underline{a}}^{\infty} \frac{\partial V(a, n, z, g(a', n_j); \tilde{\mu}^g)}{\partial f} \tilde{\mu}^g(a', n_j; z, g) da' \right\} \end{aligned}$$

$$\text{subject to } \partial_a V(\underline{a}, n, z, g; \tilde{\mu}^g) \geq u'(w(z, g)n + r(z, g)\underline{a})$$

and the optimal consumption and saving policy functions satisfy

$$u' [c^*(a, n, z, g; \tilde{\mu}^g)] = \partial_a V(a, n, z, g; \tilde{\mu}^g)$$

$$s(a, n, z, g; \tilde{\mu}^g) = w(z, g)n + r(z, g)a - c^*(a, n, z, g; \tilde{\mu}^g).$$

ii) Firm Optimality and Market Clearing

The pricing functions satisfy the equations

$$r(z, g) = \alpha e^z \left(\sum_{j=1}^2 \int_0^1 a g(a, n_j) da \right)^{\alpha-1} \left(\sum_{j=1}^2 n_j \int_0^1 g(a, n_j) da \right)^{1-\alpha} - \delta$$

$$w(z, g) = (1 - \alpha) e^z \left(\sum_{j=1}^2 \int_0^1 a g(a, n_j) da \right)^{\alpha} \left(\sum_{j=1}^2 n_j \int_0^1 g(a, n_j) da \right)^{-\alpha}.$$

iii) **Belief Consistency**

Letting the KFE be

$$dg_t(a, n) = \underbrace{\left(-\partial_a [\mathbf{s}(a, n, Z_t, g_t; \tilde{\mu}^g) g_t(a, n)] + \lambda(\check{n}) g_t(a, \check{n}) - \lambda(n) g_t(a, n) \right)}_{\mu^g(a, n; Z_t, g_t | \tilde{\mu}^g)} dt,$$

we have

$$\mu^g(\cdot; z, g) := \mu^g(\cdot; z, g | \mu^g) = \tilde{\mu}^g(\cdot; z, g).$$

3.5 The Master Equation

3.5.1 With a Borrowing Constraint

Defining $V(\cdot) = V(\cdot; \mu^g)$ and directly baking belief consistency into the household HJBE gives us the master equation

$$\rho V(a, n, z, g) = u \left((u')^{-1} [\partial_a V(a, n, z, g)] \right) \quad (\text{Optimal Consumption})$$

$$+ \partial_a V(a, n, z, g) \left(w(z, g)n + r(z, g)a - (u')^{-1} [\partial_a V(a, n, z, g)] \right) \quad (\text{Individual Wealth})$$

$$+ \lambda(n) \left(V(a, \check{n}, z, g) - V(a, n, z, g) \right) \quad (\text{Idiosyncratic Shock})$$

$$+ \partial_z V(a, n, z, g) \mu_z(z) + \frac{1}{2} \sigma^2 \partial_{zz} V(a, n, z, g) \quad (\text{Aggregate Shock})$$

$$+ \sum_{j=1}^2 \int_{\underline{a}}^{\infty} \frac{\partial V(a, n, z, g(a', n_j))}{\partial f} \mu^g(a', n_j; z, g) da' \quad (\text{Belief Consistency + Wealth Distribution})$$

$$\text{subject to } \partial_a V(\underline{a}, n, z, g) \geq u'(w(z, g)n + r(z, g)\underline{a}) \quad (\text{Boundary Condition})$$

$$\begin{aligned} \mu^g(a, n; z, g) = & -\partial_a \left[\left(w(z, g)n + r(z, g)a - (u')^{-1} [\partial_a V(a, n, z, g)] \right) g(a, n) \right] \\ & + \lambda(\check{n}) g(a, \check{n}) - \lambda(n) g(a, n) \quad (\text{Drift of KFE}) \end{aligned}$$

$$r(z, g) = \alpha e^z \left(\sum_{j=1}^2 \int_0^1 a g(a, n_j) da \right)^{\alpha-1} \left(\sum_{j=1}^2 n_j \int_0^1 g(a, n_j) da \right)^{1-\alpha} - \delta \quad (\text{Interest Rate})$$

$$w(z, g) = (1 - \alpha) e^z \left(\sum_{j=1}^2 \int_0^1 a g(a, n_j) da \right)^{\alpha} \left(\sum_{j=1}^2 n_j \int_0^1 g(a, n_j) da \right)^{-\alpha}. \quad (\text{Wage})$$

3.5.2 With a Soft Borrowing Penalty

With a soft penalty $\psi(a)I_{\{a \leq \underline{a}\}}$ instead of a hard borrowing constraint, the state space for wealth becomes $\mathcal{A} = \mathbb{R}$ and the master equation in terms of the derivative $W(a, n, z, g) = \partial_a V(a, n, z, g)$ is

$$0 = (r(z, g) - \rho)W(a, n, z, g) + \psi'(a)I_{\{a \leq \underline{a}\}} \quad (\text{Optimal Consumption})$$

$$+ \partial_a W(a, n, z, g) \left(w(z, g)n + r(z, g)a - (u')^{-1} [W(a, n, z, g)] \right) \quad (\text{Individual Wealth})$$

$$+ \lambda(n) \left(W(a, \check{n}, z, g) - W(a, n, z, g) \right) \quad (\text{Idiosyncratic Shock})$$

$$+ \partial_z W(a, n, z, g) \eta(\bar{Z} - z) + \frac{1}{2} \sigma^2 \partial_{zz} W(a, n, z, g) \quad (\text{Aggregate Shock})$$

$$+ \sum_{j=1}^2 \int_{-\infty}^{\infty} \frac{\partial W(a, n, z, g(a', n_j))}{\partial f} \mu^g(a', n_j; z, g) da' \quad (\text{Belief Consistency + Wealth Distribution})$$

subject to $\mu^g(a, n; z, g) = -\partial_a \left[\left(w(z, g)n + r(z, g)a - (u')^{-1} [W(a, n, z, g)] \right) g(a, n) \right] + \lambda(\check{n})g(a, \check{n}) - \lambda(n)g(a, n) \quad (\text{Drift of KFE})$

$$r(z, g) = \alpha e^z \left(\sum_{j=1}^2 \int_0^1 a g(a, n_j) da \right)^{\alpha-1} \left(\sum_{j=1}^2 n_j \int_0^1 g(a, n_j) da \right)^{1-\alpha} - \delta \quad (\text{Interest Rate})$$

$$w(z, g) = (1 - \alpha) e^z \left(\sum_{j=1}^2 \int_0^1 a g(a, n_j) da \right)^{\alpha} \left(\sum_{j=1}^2 n_j \int_0^1 g(a, n_j) da \right)^{-\alpha}. \quad (\text{Wage})$$

3.6 Neural Net Implementation

Given the theoretical results derived so far, here I provide some exposition about the details that goes into my version of solving to the master equation via a finite agent approximation and a soft borrowing penalty via deep learning.

3.6.1 A Finite Agent Approximation

We approximate the wealth distribution using a finite number of agents. Suppose there are $N + 1$ agents in the economy and that each household acts as a price taker: then, the value function of household i now depends on the wealth and labor supply of the remaining N agents $\varphi^{-i} = \{a_j, n_j\}_{1 \leq j \leq N}$ since the interest rate and wage are given as functions of the aggregate shock z and the asset/labor supply positions φ^{-i} of the other agents:

$$r(z, \varphi^{-i}) = \alpha e^z \left(\frac{1}{N} \sum_{j=1}^N a_j \right)^{\alpha-1} \left(\frac{1}{N} \sum_{j=1}^N n_j \right)^{1-\alpha} - \delta$$

$$w(z, \varphi^{-i}) = (1 - \alpha) e^z \left(\frac{1}{N} \sum_{j=1}^N a_j \right)^{\alpha} \left(\frac{1}{N} \sum_{j=1}^N n_j \right)^{-\alpha}.$$

The master equation with a soft borrowing penalty now becomes

$$\begin{aligned} 0 = \mathcal{L}W(a_i, n_i, z, \varphi^{-i}) &:= (r(z, \varphi^{-i}) - \rho)W(a_i, n_i, z, \varphi^{-i}) + \psi'(a_i) \cdot I_{\{a_i \leq \underline{a}\}} \\ &+ \partial_{a_i} W(a_i, n_i, z, \varphi^{-i}) \left(w(z, \varphi^{-i}) n_i + r(z, \varphi^{-i}) a_i - (u')^{-1} [W(a_i, n_i, z, \varphi^{-i})] \right) \\ &+ \lambda(n_i) \left(W(a_i, \check{n}_i, z, \varphi^{-i}) - W(a_i, n_i, z, \varphi^{-i}) \right) \\ &+ \partial_z W(a_i, n_i, z, \varphi^{-i}) \eta(\bar{Z} - z) + \frac{1}{2} \sigma^2 \partial_{zz} W(a_i, n_i, z, \varphi^{-i}) \\ &+ \sum_{j=1}^N \left[\partial_{a_j} W(a_i, n_i, z, \varphi^{-i}) \left(w(z, \varphi^{-j}) n_j + r(z, \varphi^{-j}) a_j - (u')^{-1} [W(a_i, n_i, z, \varphi^{-i})] \right) \right. \\ &\quad \left. + \lambda(n_j) \left(W(a_i, n_i, z, \{\{a_{j'}, n_{j'}\}_{j' \neq j, i}, (a_j, \check{n}_j)\}) - W(a_i, n_i, z, \varphi^{-i}) \right) \right] \end{aligned}$$

The term

$$\partial_{a_j} W(a_i, n_i, z, \varphi^{-i}) \left(w(z, \varphi^{-j}) n_j + r(z, \varphi^{-j}) a_j - (u')^{-1} [W(a_i, n_i, z, \varphi^{-i})] \right)$$

is the change in household i 's value function given the change in the savings of household j , while

$$\lambda(n_j) \left(W(a_i, n_i, z, \{\{a_{j'}, n_{j'}\}_{j' \neq j, i}, (a_j, \check{n}_j)\}) - W(a_i, n_i, z, \varphi^{-i}) \right)$$

is the change in household i 's value function given the change in the employment status of household j .

3.6.2 Shape Constraints and the MPC

In practice, alongside a loss function involving $\mathcal{L}W(a_i, n_i, z, \varphi^{-i})$, we need to impose additional shape constraints that ensure the resulting value function is concave. Specifically, we need to impose the concavity constraints

$$\partial_{a_i} W(a_i, n_i, z, \varphi^{-i}), \partial_z W(a_i, n_i, z, \varphi^{-i}) < 0.$$

The first constraint in particular warrants some attention, as it is intimately to the marginal propensity to consume.

Returning briefly to the original problem with density g , under CRRA utility with RRA coefficient γ , optimal consumption is given as

$$c^*(a, n, z, g) = W(a, n, z, g)^{-\frac{1}{\gamma}}.$$

As such, the marginal propensity to consume is

$$\frac{\partial c^*(a, n, z, g)}{\partial a} = -\frac{1}{\gamma} \partial_a W(a, n, z, g) (W(a, n, z, g))^{-\frac{\gamma+1}{\gamma}}.$$

For standard models with heterogeneous households, we know that:

- The MPC for households with large asset holdings is generally lower than those with asset holdings near the borrowing constraint
- The MPC for employed households is generally lower than that of unemployed households (since the borrowing of these households are more likely to be constrained)
- The asset grid is chosen so that households with asset holdings near the upper boundary of the asset grid should dissave.
- Households have a target level of wealth. As such, the asset grid is chosen so that households with asset holdings near the upper boundary of the asset grid dissave, and the degree of dissaving is larger for households with greater wealth.

We want to refine the concavity constraint so that these four features of the problem are satisfied. To this end, we think of the consumption decisions of employed households that hold

assets in an $h > 0$ neighborhood around the upper bound of the asset space a_{max} . We know that

$$\begin{aligned} c^*(a_{max} - h, n_2, z, g) - w(z, g)n_2 - r(z, g)(a_{max} - h) &\geq 0 \\ c^*(a_{max}, n_2, z, g) - w(z, g)n_2 - r(z, g)a_{max} &\geq 0 \end{aligned}$$

by dissaving. Furthermore,

$$c^*(a_{max}, n_2, z, g) - w(z, g)n_2 - r(z, g)a_{max} \geq c^*(a_{max} - h, n_2, z, g) - w(z, g)n_2 - r(z, g)(a_{max} - h) \geq 0$$

because households with more wealth dissave more near the upper bound of the asset space. It follows that

$$\frac{c^*(a_{max} - h, n_2, z, g) - c^*(a_{max}, n_2, z, g)}{-h} \geq r(z, g),$$

and taking $h \rightarrow 0$ on both sides shows us that

$$\frac{\partial c^*(a_{max}, n_2, z, g)}{\partial a} \geq r(z, g).$$

Finally, since

$$\frac{\partial c^*(a, n, z, g)}{\partial a} \geq \frac{\partial c^*(a_{max}, n_2, z, g)}{\partial a} \geq r(z, g),$$

it must be the case that the MPC given the aggregate states is always bounded below by the interest rate given those aggregate states. This imposes the constraint

$$-\frac{1}{\gamma} \partial_a W(a, n, z, g) (W(a, n, z, g))^{-\frac{\gamma+1}{\gamma}} \geq r(z, g)$$

on $W(\cdot)$, so our concavity constraint is therefore given as the mean of

$$\max \left[r(z, g) + \frac{1}{\gamma} \partial_a W(a, n, z, g) (W(a, n, z, g))^{-\frac{\gamma+1}{\gamma}}, 0 \right]^2$$

over the sample points.

In the context with finite agents, our shape constraint is therefore a function of

$$\begin{aligned} \mathcal{E}^s(a_i, n_i, z, \varphi^{-i}) &= \max \left[r(z, \varphi^{-i}) + \frac{1}{\gamma} \partial_a W(a_i, n_i, z, \varphi^{-i}) (W(a_i, n_i, z, \varphi^{-i}))^{-\frac{\gamma+1}{\gamma}}, 0 \right]^2 \\ &\quad + \max \left[\partial_z W(a_i, n_i, z, \varphi^{-i}), 0 \right]^2. \end{aligned}$$

At each epoch, given a sample $\{a_i^{(l)}, n_i^{(l)}, z^{(l)}, \varphi^{-i(l)}\}_{1 \leq l \leq M}$ of size M , the loss at that epoch

is given as

$$\mathcal{E} = \kappa^e \frac{1}{M} \sum_{l=1}^M \left| \mathcal{LW}(a_i^{(l)}, n_i^{(l)}, z^{(l)}, \varphi^{-i(l)}) \right|^2 + \kappa^s \frac{1}{M} \sum_{l=1}^M \mathcal{E}^s(a_i^{(l)}, n_i^{(l)}, z^{(l)}, \varphi^{-i(l)}),$$

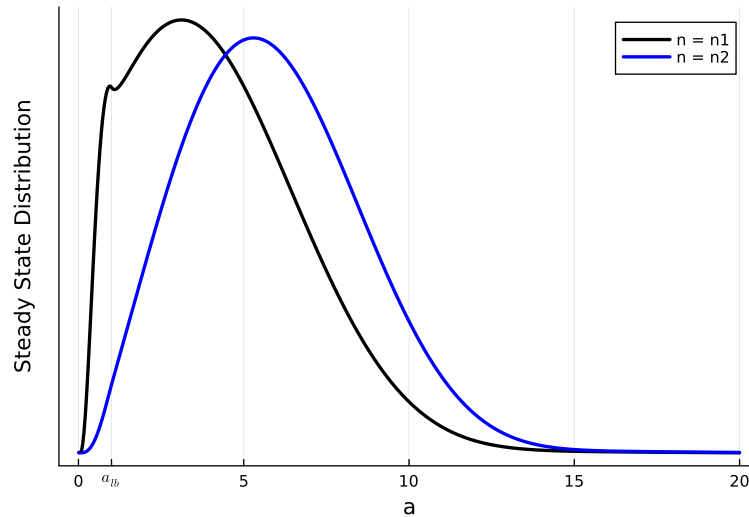
where κ^e and κ^s are weights on each of the components of the loss.

3.6.3 Sampling Procedure

At each epoch we sample M points $\{a_i^{(l)}, n_i^{(l)}, z^{(l)}, \{a_j^{(l)}, n_j^{(l)}\}_{j \neq i}\}_{1 \leq l \leq M}$ at random. Instead of active sampling for $a_i^{(l)}$, I tried an alternative sampling scheme that would sample points from the asset space $\mathcal{A}$ in accordance to what would likely be the wealth distribution given the aggregate shocks.

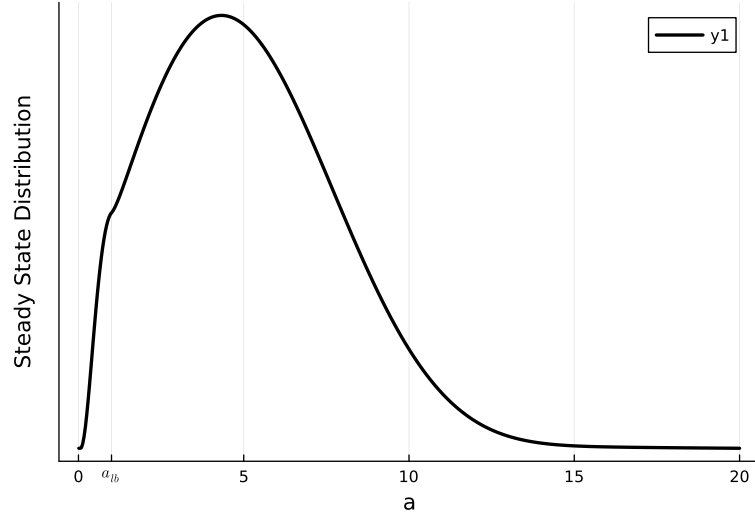
Instead of sampling household i 's wealth $a_i^{(l)}$ from the uniform distribution on $[a_{min}, a_{max}]$, I wanted to sample it from a distribution that allows us to more heavily sample from areas where households are likely to hold asset positions. To this end, I looked for a flexible parametric approximation to the steady state wealth distribution.

Estimating the steady state distribution via the finite difference method shows us that the stationary distribution $g(a, n)$ at the steady state with log aggregate productivity equal to 0 is given as



The steady state wealth distribution $g_{ss}(a) = g_{ss}(a, n_1) + g_{ss}(a, n_2)$ then becomes

The intuition for my sampling schemes comes from the observation that this distribution looks very much like your usual bell-shaped distribution. My first instinct was to approximate the steady state distribution using a normal distribution truncated to $[a_{min}, a_{max}]$ with the same mean and variance as the steady state distribution, but this requires numerical methods to solve for the requisite parameters of the truncated normal distribution that would immensely slow



down the sampling process.

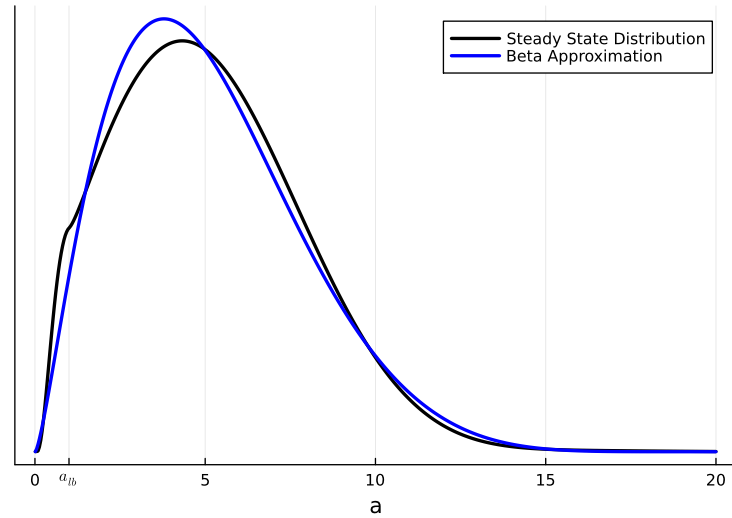
My second thought was to restrict my attention to known bell-shaped distributions with compact support; and a natural candidate was a rescaling and translation of the beta distribution. Specifically, we restrict our attention to the distribution of the random variable

$$\mathbf{G} = (a_{max} - a_{min})\beta + a_{min}, \quad \beta \sim \mathcal{B}(\alpha, \beta),$$

where α, β are chosen so that the mean and variance of $\mathbf{G}$ are exactly those of the steady state distribution:

$$\frac{\alpha}{\alpha + \beta} = k_{ss}, \quad \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = V_a.$$

This approximate works surprisingly better than expected, as we see in the plot below:



This suggests a natural sampling scheme for $\{a_i^{(l)}, n_i^{(l)}, z^{(l)}, \{a_j^{(l)}, n_j^{(l)}\}_{j \neq i}\}_{1 \leq l \leq M}$. I sample the l th point $(a_i^{(l)}, n_i^{(l)}, z^{(l)}, \{a_j^{(l)}, n_j^{(l)}\})$ as follows:

- $n_i^{(l)}$ is sampled from $\{n_1, n_2\}$ according to their steady state probabilities (which in our case is 0.5 for each point).

We sample $\{n_j^{(l)}\}_{j \neq i}$ from $\{n_1, n_2\}$ just like we sample $n_i^{(l)}$, using which we obtain the implied labor supply as

$$\bar{L}^{(l)} = \frac{1}{N} \sum_{j \neq i} n_j^{(l)}.$$

- $z^{(l)}$ is sampled uniformly from the range $[z_{min}, z_{max}]$
- We sample the desired interest rate r uniformly from $[\rho - 0.1, \rho]$. The implied level of capital that would be demanded under this interest rate is computed as

$$k(r) = \left(\frac{\alpha e^{z^{(l)}} (\bar{L}^{(l)})^{1-\alpha}}{r + \delta} \right)^{\frac{1}{1-\alpha}}.$$

- $a_i^{(l)}$ is sampled from the distribution of

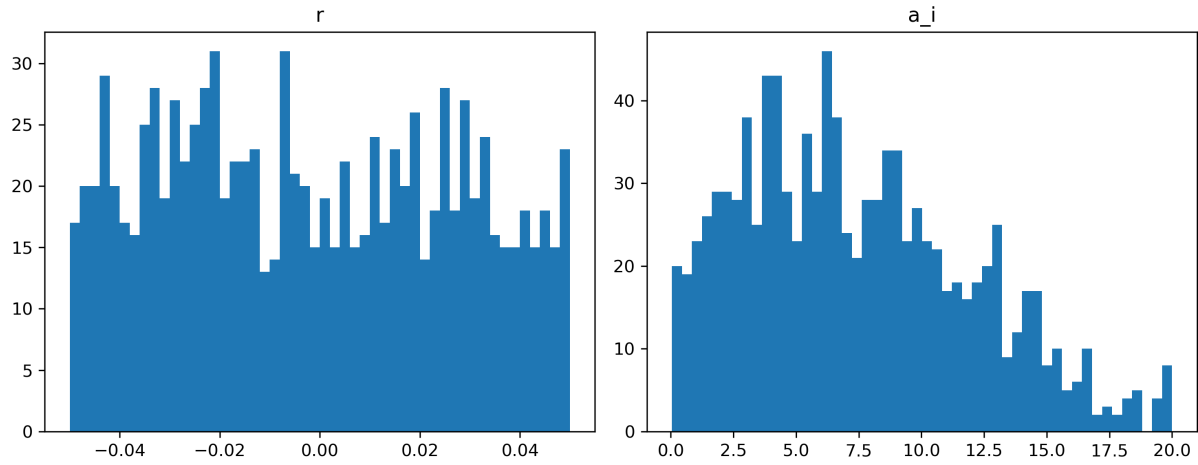
$$\mathbf{G}_{ss} = (a_{max} - a_{min})\beta(r) + a_{min}, \quad \beta_{ss} \sim \mathcal{B}(\alpha(r), \beta(r)),$$

where $\alpha(r)$ and $\beta(r)$ are chosen to ensure that the mean of $\mathbf{G}(r)$ is $k(r)$ and its variance is the steady state variance V_a .

- Finally, we sample $\{a_j^{(l)}\}_{j \neq i}$ from the uniform distribution on $[a_{min}, a_{max}]$ and normalize the points by $\frac{k(r)}{\frac{1}{N} \sum_{j=1}^N a_j^{(l)}}$.

Below is an example of the sampled interest rates from $[-0.05, 0.05]$ and the corresponding samples of a_i :

Note how, in contrast to the uniformity of the interest rates, a_i is sampled primarily from low asset levels which is where most households hold their wealth positions.



3.6.4 Initialization

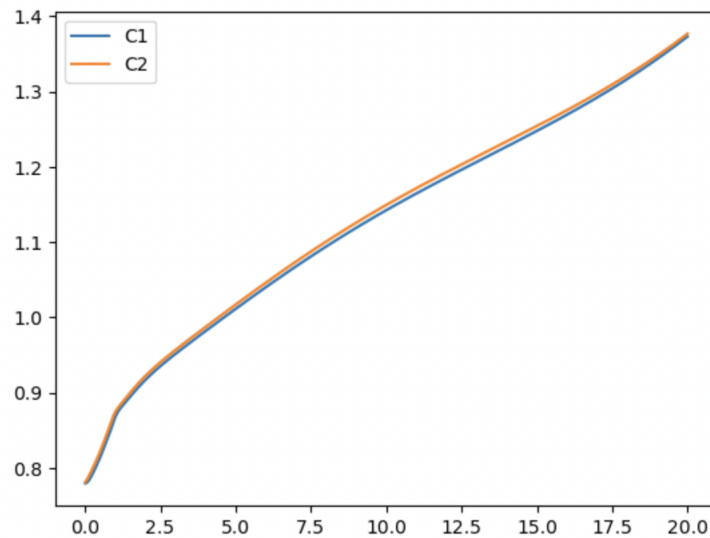
I initialize the neural network at the function

$$W(a, n, z, g) = \log(a_{max}) + 10^{-4} - \log(a),$$

which is bounded below by $1e - 4$, is downward sloping and convex, and does not decay as fast as the mapping $a \mapsto e^{-a}$ (which prevents consumption from exploding).

3.6.5 Results for Steady State Consumption

I'm running the training loop right now; it's on its 120,000th epoch with an average pde residual loss of around 0.0005. I've set the loop to terminate once this quantity reaches $1e-5$, which I suspect will take a couple hundred thousand more epochs. Right now the consumption functions look like



I expect optimal consumption to start to differ more across employment status as the training continues, but right now I'm not holding out much hope. So far this is my biggest roadblock.